

STEVIE BELLIS

Site Reliability Engineer · Systems Engineer · Protocol & Infrastructure Architect
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SUMMARY

SRE/Systems Engineer with 7 years designing & operating hybrid infrastructure and AWS for enterprise SaaS (10 years total IT). Currently building Unheaded, an open-source infrastructure-as-code platform (500K+ LOC, Go/Rust, GPL-3.0) featuring a custom IPv6 protocol suite with six IETF Internet-Draft specifications, event streaming at 920K events/sec, eBPF packet-level telemetry, and a protocol-native virtual CPU architecture. Production expertise in Linux systems engineering, infrastructure automation (Puppet/Ruby IaC across 1,000+ systems), networking (BGP, OSPF, EVPN, VXLAN, MPLS), database operations (PostgreSQL, Oracle, Redis), and incident response with sub-hour MTTR. Technical range spanning kernel tuning to cross-connect turn-ups, with experience building high-throughput streaming pipelines and data infrastructure orchestration for large-scale distributed systems.

KEY ACHIEVEMENTS

- Authored six IETF Internet-Draft protocol specifications (wire format, BPF dictionary, memory model, eBPF pipeline, extensions) with 12+ IANA registry proposals for a novel IPv6 Hop-by-Hop infrastructure protocol suite.
- Designed data pipeline and event-driven architecture with ring buffer message bus sustaining 920K events/sec, zero drops, topic-based pub/sub with at-least-once delivery, and full end-to-end observability.
- Deployed multi-datacenter BGP EVPN/VXLAN fabric (Juniper MX/QFX/EX + FRR/BIRD) with 10+ Gbps throughput across 1,000+ systems at 99.99%+ uptime.
- Scripted migration of 50+ production databases (~3–5 TB) to AWS RDS Multi-AZ with zero data loss; built autonomous validation, cutover, and rollback automation.
- Automated customer provisioning (Puppet + Ruby + Bash) cutting onboarding from hours to minutes; maintained SOC 2 compliance baselines across 1,000+ systems.
- Reduced AWS spend ~30% via Reserved Instances, Savings Plans, and lifecycle policies while maintaining performance SLAs.

CORE SKILLS

Networking: IPv6, BGP, EVPN, VXLAN, OSPF, IS-IS, MPLS, QUIC/HTTP/3, gRPC, mTLS, WireGuard; Juniper MX/EX/QFX, FRR/BIRD
IaC & CI/CD: Puppet (Ruby modules), Git, Jenkins, GitHub Actions; familiarity with Terraform, Ansible, NixOS
Systems: Linux (Rocky/RHEL, Ubuntu, NixOS), eBPF (XDP/TC/kprobe), kernel tuning, cgroups v2, systemd
Cloud: AWS (VPC, EC2, RDS, IAM, CloudWatch, Direct Connect, S3, Multi-AZ)
Containers: Docker, LXC/LXD, containerd; familiarity with Kubernetes, OCI runtimes
Observability: Grafana, Prometheus, Loki, VictoriaMetrics, PagerDuty, custom eBPF telemetry pipelines
Security: SOC 2 Type II, NIST/CIS/STIG, Zero Trust, mTLS, MITRE ATT&CK, IAM/SSO (SAML/OAuth), OpenLDAP, threat modeling, red/blue team
Data Center: Racking, copper/fiber cabling, hot/cold aisle, cross-connect/MMR workflows, Direct Connect turn-ups
Data: PostgreSQL, Oracle (ops/migrations), Redis, event streaming (ring buffers, pub/sub, Apache Flume), CBOR serialization
Languages: Go, Rust, Ruby, Python, Bash; familiarity with Java, C, JavaScript

EXPERIENCE

Creator & Lead Engineer | *Unheaded (Open Source, GPL-3.0)* Jan 2026 – Present

Open-source IaC platform: 25+ microservices, custom IPv6 protocol suite, eBPF telemetry, virtual CPU. 500K+ LOC in Go and Rust.

- Architected 4-tier microservice platform with gRPC service mesh, circuit breakers, L4/L7 load balancing (Maglev hashing), and declarative state reconciliation (desired vs. actual).
- Designed custom IPv6 Hop-by-Hop protocol: 20-byte per-packet register with CRC-16/CCITT integrity verification, exponent encoding, and BPF map dictionaries for service/action semantics.
- Built eBPF telemetry pipeline (Rust/Aya): XDP packet capture → ring buffer → event processing → Prometheus/Grafana dashboards with zero-copy kernel-to-userspace tracing.
- Engineered protocol-native virtual CPU (UPC) with custom bytecode ISA (MBC), 5-level memory hierarchy, and eBPF execution bridge.
- Operated bare metal lab: NixOS fleet, BGP routing (FRR/BIRD), EVPN-VXLAN overlay, Redis, PostgreSQL, Docker/LXD containers, WireGuard mesh.

Systems Engineer / SRE | *Supplyframe (acquired by Siemens)* Dec 2017 – Apr 2025

Cybersecurity Professional (internal transfer Oct 2024). Shared platform infra: hybrid DCs + AWS, 1,000+ systems, enterprise SaaS.

- Authored Puppet modules (Ruby) for service deployment, HAProxy/Nginx load balancers, and VPC provisioning; maintained IaC across 1,000+ systems with Git-driven change control.
- Built and operated multi-datacenter BGP EVPN/VXLAN fabric (Juniper MX/QFX/EX + FRR/BIRD) and hybrid connectivity via AWS Direct Connect with 10+ Gbps throughput.
- Scripted 50+ database migrations (~3–5 TB) to AWS RDS Multi-AZ — autonomous validation, phased cutover, rollback automation, zero data loss.
- Managed OpenLDAP as centralized auth; deployed tenant-isolated LDAP instances with custom OIDs for fine-grained identity attributes and authorization logic.
- Owned 24/7 on-call with sub-hour MTTR: incident response, runbooks, blameless post-incident reviews; drove prevention-focused durable fixes.
- Partnered with red/blue teams on security assessments and hardening; enforced Zero Trust, IAM/SAML/OAuth; drove SOC 2 Type II first-time certification with automated evidence collection.
- Drove AWS cost governance (RI/Savings Plans/lifecycle) reducing spend ~30%; mentored junior engineers; participated in architectural design reviews.
- Physical DC work: racked/cabled servers and network hardware; coordinated cross-connect and Direct Connect turn-ups with remote hands.

Network Admin / Desktop Support | *Leif Johnson Ford* Jan 2014 – Dec 2017

- Administered network/systems infrastructure across 6 campuses (375 users, 700 devices): Cisco ASA, SonicWall VPN/IPS, SNMP monitoring, Linux/Windows server environments.

EDUCATION AND CERTIFICATIONS

Austin Community College — Coursework towards UT Computer Science Transfer program	Fall 2017
Cisco Certified Network Associate (CCNA)	January 2017–2020
CompTIA A+ Certification	2011
Jefferson Community and Technical College — Coursework towards Computer Information Systems Associates Degree	2008–2009
Oldham County Career Center — Information Technology Vocational Certificate(A+, CCNA, Visual Basic .NET/VB.NET)	2004–2006